



BAULKHAM HILLS HIGH SCHOOL

TRIAL 2013
YEAR 12 TASK 4

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 180 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-16
- Marks may be deducted for careless or badly arranged work

Total marks – 100

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks)
Questions 1-10

- Attempt Question 1-10
- Allow about 15 minutes for this section

Section II – Pages 5-10 (90 marks)

- Attempt questions 11-16
- Allow about 2 hours 45 minutes for this section

Table of Standard Integrals is on page 11

Section I - 10 marks**Allow about 15 minutes for this section****Use the multiple choice answer sheet for question 1-10**

1. Which of the following is equal to $\cos \theta$?

(A) $\frac{\sin \theta}{\tan \theta}$

(B) $\sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}$

(C) $2 \cos^2 \theta - 1$

(D) $2 \cos^2 \frac{\theta}{2} + 1$

2. In Cartesian form $2 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$ is

(A) $-\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$

(B) $-i$

(C) $-\sqrt{2}(1 - i)$

(D) $\sqrt{2}(1 - i)$

3. Using an appropriate substitution

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{(1 + \tan x)^2} dx$ is equivalent to:

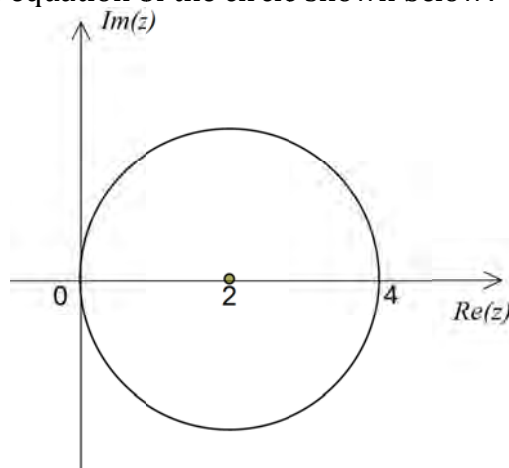
(A) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u} du$

(B) $\int_0^2 \frac{u^2}{(1 + u)^3} du$

(C) $\int_0^2 \frac{1}{u^2} du$

(D) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u^3} du$

4. Which of the following is the equation of the circle shown below?



- (A) $(z + 2)(\bar{z} + 2) = 4$
 (B) $(z - 2)(\bar{z} - 2) = 4$
 (C) $(z + 2i)(\bar{z} - 2i) = 4$
 (D) $(z + 2)(\bar{z} - 2) = 4$

5. Using implicit differentiation on the equation $y^3 = x^2 + xy$, then $\frac{dy}{dx}$ would equal

- (A) $\frac{3y^2 - 2x}{x}$
 (B) $\frac{2x + y}{3y^2 - x}$
 (C) $\frac{2x - y}{3y^2 + y}$
 (D) $\frac{2x}{3y^2 + y}$

6. A satellite in a circular orbit around Earth, at a distance of 12000 km from Earth's centre makes 12 revolutions per day. Find the tangential speed of the satellite in km/h.

- (A) π
 (B) $\frac{72000}{\pi}$
 (C) 12000π
 (D) $12000\pi^2$

7.	If α, β and γ are the roots of the equation $x^3 - 3x + 4 = 0$ Then the cubic with roots α^2, β^2 and γ^2 is (A) $8x^3 - 9x + 4 = 0$ (B) $x^3 - 6x^2 + 9x - 16 = 0$ (C) $x^3 + 9x^2 - 12x + 4 = 0$ (D) $8x^3 + 4x^2 - 9x + 16 = 0$
8.	Given $(2i + 1)$ is a root of the equation $x^3 - 4x^2 + 9x - 10 = 0$ then another root is (A) 2 (B) 5 (C) $2i - 1$ (D) 10
9.	$\tan(\cos^{-1} x)$ is equal to (A) $-\frac{\sqrt{1-x^2}}{x}$ (B) $-\frac{x}{\sqrt{1-x^2}}$ (C) $\frac{\sqrt{1-x^2}}{x}$ (D) $\frac{x}{\sqrt{1-x^2}}$
10.	$\int x\sqrt{1-x} dx$ equals (A) $-\frac{1}{3}x^2(1-x)^{\frac{3}{2}} + c$ (B) $\frac{1}{3}x^2(1-x)^{\frac{3}{2}} + c$ (C) $-\frac{2}{5}x(1-x)^{\frac{5}{2}} + \frac{2}{3}(1-x)^{\frac{3}{2}} + c$ (D) $\frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + c$
End of Section 1	

Section II – Extended Response

Attempt questions 11-16.

Answer each question on a SEPARATE PAGE. Clearly indicate question number.

Each piece of paper must show your name.

All necessary working should be shown in every question.

Question 11 (15 marks)		Marks
a)	Let $z_1 = 3 - 4i$ and $z_2 = -3 + 2i$	
	(i) $z_1 - \bar{z}_2$	1
	(ii) $\frac{z_1}{z_2}$	2
b)	Given that $(1 - 2i)^2 = -3 - 4i$, solve $z^2 - 5z + (7 + i) = 0$.	2
c)	On an Argand diagram, shade the region specified by the conditions $ z - 6 + 5i \leq 3$ and $\operatorname{Re}(z) \leq 6$.	2
d)	If $z = a(\cos \theta + i \sin \theta)$ when a and θ are real, show that $\frac{z}{z^2 + a^2}$ is equivalent to $\frac{1}{2a \cos \theta}$	3
e)	(i) Prove that if $y = (x + \sqrt{1 + x^2})^m$ then $\frac{1}{y} \cdot \frac{dy}{dx} = \frac{m}{\sqrt{1 + x^2}}$	2
	(ii) Show $\frac{d^2y}{dx^2} = \frac{m^2 y \sqrt{1 + x^2} - m y x}{(1 + x^2) \sqrt{1 + x^2}}$	2
	(iii) Prove that $(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - m^2 y = 0$	1
End of Question 11		

Question 12 (15 marks)		Marks
a)	Find $\int \frac{dx}{(x+1)(x^2+2)}$	3
b)	(i) Show that $\log_{ab} x = \frac{\log_a x}{1+\log_a b}$	2
	(ii) Hence show that $\log_2 5 = \frac{1-\log_{10} 2}{\log_{10} 2}$	1
c)	Consider the curves $\frac{x^2}{16} + \frac{y^2}{7} = 1 \quad \text{and} \quad x^2 - \frac{y^2}{8} = 1$	
	(i) Show that both curves have the same foci.	3
	(ii) Find the equation of the circle that passes through the points of intersection of these two curves.	3
d)	(i) In how many distinct ways can the letters of the word ANGLE be arranged.	1
	(ii) If these arrangements are listed in alphabetical order, in which place (ie. 1 st , 2 nd , 3 rd , etc...) is the word ANGLE .	2
End of Question 12		

Question 13 (15 marks)	Marks
a) $I_n = \int_0^{\pi} \sin^n x \, dx$ (i) Prove that $I_n = \frac{n-1}{n} I_{n-2}$ (ii) Hence evaluate I_5	3 2
b) When a polynomial $P(x)$ is divided by $(x - 3)$ and $(x - 7)$ the respective remainders are 3 and 5. Find the remainder when $P(x)$ is divided by $(x - 3)(x - 7)$.	3
c) Two circles of equal radii intersect at A and B. X is a point on the circle between A and B and BX is produced to meet the second circle at Y. Copy the diagram in your booklet and prove that $AX = AY$, showing any necessary constructions.	3
d) Find the volume of the solid of revolution generated when the area enclosed between the curve $y = 4 - x^2$ and the lines $y = 4$ and $x = 2$ is rotated about the line $x = 2$.	4
End of Question 13	

Question 14 (15 marks)	Marks
a) A body of unit mass falls under gravity through a resistive medium. The body falls from rest from a cliff 50 metres above the ground. The resistance to its motion is $\frac{v^2}{100}$ where $v \text{ m s}^{-1}$ is the speed of the body when it has fallen a distance of x metres. (i) Show that the equation of the motion is $\ddot{x} = g - \frac{v^2}{100}$ (ii) Show that the terminal velocity V of the body is given by $V = 10\sqrt{g} \text{ ms}^{-1}$ (iii) Show that $v^2 = V^2 \left(1 - e^{-\frac{x}{50}}\right)$. (iv) How far has the body fallen when it reaches a velocity of $\frac{V}{2}$. (v) Find the velocity reached in terms of the terminal velocity when the body hits the ground. (vi) If $v = v_1$ when $x = d$ and $v = v_2$ when $x = 2d$, show that $v_2^2 = v_1^2 \left(2 - \frac{v_1^2}{V^2}\right)$	1 1 3 2 2 2
b) The equation $x^4 - 5x^3 - 9x^2 + ax + b = 0$ has a triple root. Given that this root is an integer: (i) find the triple root. (ii) find the value of b.	2 2
End of Question 14	

Question 15 (15 marks)	Marks
a) (i) Prove by mathematical induction that $(1 + x)^n - 1$ is divisible by x for all integers $n \geq 1$. (ii) By factorising $35^n - 7^n - 5^n + 1$ and using part (i), prove that $35^n - 7^n - 5^n + 1$ is divisible by 24.	3 2
b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 4 \cos x}$	4
c) The base of a certain solid is the region bounded by the curve $y^2 = 4ax$ and $x^2 = 4ay$ and cross sections to the plane perpendicular to the x-axis are semi circles. (i) Show that the two curves intersect at $A(0,0)$ and $B(4a, 4a)$. (ii) Show that the cross sectional area, A, of a typical slice is $A = \frac{\pi}{2} \left(\sqrt{ax} - \frac{x^2}{8a} \right)^2$. (iii) Hence find the volume of the solid formed.	1 2 3
End of Question 15	

Question 16 (15 marks)		
a)	P is a point $\left(p, \frac{1}{p}\right)$ on the rectangular hyperbola $xy = 1$. The line PO is produced to point Q also on the rectangular hyperbola. A circle centre P and radius PQ is drawn to cut the hyperbola at A, B, C and Q. (i) Prove that the parameters of the points of intersection of the circle and the hyperbola are given by the equation $p^2 t^4 - 2p^3 t^3 - 3(p^4 + 1)t^2 - 2pt + p^2 = 0$ (ii) Deduce that $t_A + t_B + t_C = 3p$ where t_A, t_B and t_C are the parameters at A, B and C	3 2
b)	(i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ where $\cos \theta \neq 0$ and n is a positive integer. (ii) Hence show that if z is a purely imaginary number, the roots of $(1 + z)^4 + (1 - z)^4 = 0$ are $z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$.	2 3
c)	Consider the sequence defined by $V_k = \frac{1}{2k+1} + \frac{1}{2k+2} + \cdots + \frac{1}{3k}$ where k is a positive integer (i) Show that $V_k < \frac{1}{2}$ (ii) Given that $p < x < p + 1$, where x is a real number and p is a positive integer show that $\frac{1}{p+1} < \int_p^{p+1} \frac{dx}{x} < \frac{1}{p}$ (iii) Hence show that $\int_{2k+1}^{3k+1} \frac{dx}{x} < V_k < \int_{2k}^{3k} \frac{dx}{x}$ (iv) Hence find the limit of V_k as $k \rightarrow \infty$	1 1 2 1
End of Paper		

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1. A 2. C 3. C 4. B 5. B
6. C 7. B 8. A 9. C 10. D

Ques 11

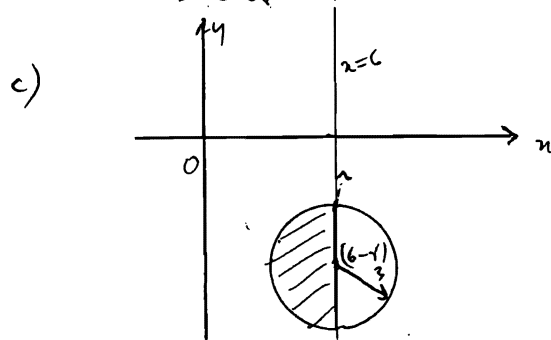
a) (i) $z_1 - \bar{z}_2$
 $= (3-4i) - (-3-2i)$
 $= 6-2i$

(ii) $\frac{z_1}{z_2} = \frac{3-4i}{-3+2i} \times \frac{-3-2i}{-3-2i}$
 $= \frac{(-9-8)+i(12-6)}{3^2+2^2}$
 $= \frac{-17}{13} + \frac{6i}{13}$

b) $z^2 - 5z + (7+i) = 0$

$z = \frac{5 \pm \sqrt{25-4(7+i)}}{2}$
 $= \frac{5 \pm \sqrt{-3-4i}}{2}$
 $= \frac{5 \pm (1-2i)}{2} \quad (\text{as } \sqrt{-3-4i} = 1-2i)$
 $= \frac{6-2i}{2} \text{ or } \frac{4+2i}{2}$

$= 3-i \text{ or } 2+i$



d) $\frac{z}{z^2+a^2} = \frac{a(\cos\theta + i\sin\theta)}{a^2(\cos\theta + i\sin\theta)^2 + a^2}$
 $= \frac{a(\cos\theta + i\sin\theta)}{a^2(\cos^2\theta - \sin^2\theta + 2i\cos\theta\sin\theta) + a^2}$
 $= \frac{1}{a} \frac{(\cos\theta + i\sin\theta)}{(\cos^2\theta - \sin^2\theta + 2i\cos\theta\sin\theta) + 1}$

for using
De Moivre's
or
expanding.

now $\cos 2\theta = 2\cos^2\theta - 1$
 $\therefore \cos 2\theta + 1 = 2\cos^2\theta$

$\frac{z}{z^2+a^2} = \frac{1}{a} \frac{(\cos\theta + i\sin\theta)}{2\cos^2\theta + 2i\cos\theta\sin\theta} - 1$
 $= \frac{1}{a} \frac{(\cos\theta + i\sin\theta)}{2\cos\theta(\cos\theta + i\sin\theta)}$
 $= \frac{1}{2a\cos\theta}$

e) i) $y = (x + \sqrt{1+x^2})^m$
 $\ln y = m \ln(x + \sqrt{1+x^2})$
 $\frac{1}{y} \frac{dy}{dx} = \frac{m}{x + \sqrt{1+x^2}} \cdot \left(1 + \frac{x}{\sqrt{1+x^2}}\right)$
 $= \left(\frac{m}{x + \sqrt{1+x^2}}\right) \left(\frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}}\right)$
 $\therefore \frac{1}{y} \frac{dy}{dx} = \frac{m}{\sqrt{1+x^2}}$

ii) $\frac{dy}{dx} = \frac{my}{\sqrt{1+x^2}}$
 $\therefore \frac{d^2y}{dx^2} = m \frac{dy}{dx} \frac{1}{\sqrt{1+x^2}} - m y \frac{x}{\sqrt{1+x^2}^3}$
 $= m \cdot \frac{my}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} - \frac{m y x}{\sqrt{1+x^2}^3}$
 $= \frac{m^2 y}{1+x^2} - \frac{m y x}{\sqrt{1+x^2}^3}$

$$= \frac{n^2 y \sqrt{1+x^2} - m y x}{\sqrt{1+x^2} (1+x^2)}$$

=

$$\begin{aligned} \text{ii) } (1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - m^2 y &= (1+x^2) \left(\frac{m^2 y \sqrt{1+x^2} - m y x}{\sqrt{1+x^2} (1+x^2)} \right) + \frac{m y x}{\sqrt{1+x^2}} - m^2 y \\ &= m^2 y - \frac{m y x}{\sqrt{1+x^2}} + \frac{m y x}{\sqrt{1+x^2}} - m^2 y \\ &= 0 \end{aligned}$$

Ques 12

$$\text{a) } \int \frac{dx}{(x+1)(x^2+1)}$$

$$\text{Let } \frac{1}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

$$a(x^2+1) + (x+1)(bx+c) = 0$$

$$x = -1 \Rightarrow$$

$$3a = 1$$

$$a = \frac{1}{3} *$$

coefficient of x^2

$$a + b = 0$$

$$\therefore b = -\frac{1}{3} *$$

$$x = 0$$

$$2a + c = 1$$

$$\therefore c = \frac{1}{3} *$$

$$\begin{aligned} I &= \frac{1}{3} \int \frac{dx}{x+1} - \frac{1}{3} \int \frac{x dx}{x^2+1} + \frac{1}{3} \int \frac{dx}{x^2+1} \\ &= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2+1) + \frac{1}{3\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C \end{aligned}$$

$$\begin{aligned} \text{b) i) } \log_{ab} x &= \frac{\log_a x}{\log_a ab} \\ &= \frac{\log_a x}{\log_a a + \log_a b} \\ &= \frac{\log_a x}{1 + \log_a b} \end{aligned}$$

$$\text{ii) Let } a=2, b=5, x=2$$

$$\therefore \log_{10} 2 = \frac{\log_2 2}{1 + \log_2 5}$$

$$1 + \log_2 5 = \frac{1}{\log_{10} 2}$$

$$\log_2 5 = \frac{1}{\log_{10} 2} - 1 = \frac{1 - \log_{10} 2}{\log_{10} 2}$$

120

1) for ellipse $a^2=16$ $b^2=7$

$$b^2 = a^2(1-e^2)$$

$$7 = 16(1-e^2)$$

$$16e^2 = 9$$

$$e = \frac{3}{4}$$

-1

focus $(\pm ae, 0)$

$$(\pm 4 \cdot \frac{3}{4}, 0)$$

$$(\pm 3, 0)$$

+

for hyper $a^2=1$ $b^2=8$

$$b^2 = a^2(e^2 - 1)$$

$$8 = e^2 - 1$$

$$e^2 = 9$$

$$e = 3$$

focus is $(\pm ae, 0)$

$$\text{i.e. } (\pm 3, 0)$$

+

$\therefore$ both focus same

11)

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \quad \text{--- (1)}$$

$$x^2 - \frac{y^2}{8} = 1 \quad \text{--- (2)}$$

$$7x^2 + 16y^2 = 16 \times 7 \quad \text{--- (3)}$$

$$8x^2 - y^2 = 8 \quad \text{--- (4)}$$

$$(3) \times 16 \Rightarrow 16 \times 8x^2 - 16y^2 = 8 \times 16 \quad \text{--- (5)}$$

$$(3) + (5) \Rightarrow (7 + 16 \times 8)x^2 = 15 \times 16$$

$$\therefore 135x^2 = 15 \times 16$$

$$9x^2 = 16$$

$$x^2 = \frac{16}{9}$$

1

subst $\rightarrow (2)$

$$\therefore \frac{y^2}{8} = \frac{16}{9} - 1$$

$$y^2 = \frac{7}{9} \times 8$$

$$= \frac{56}{9}$$

1

by adding $\odot \quad x^2 + y^2 = r^2$

$$x^2 + y^2 = \frac{16}{9} + \frac{56}{9}$$

$$= \frac{72}{9}$$

1

$$x^2 + y^2 = 8$$

d1) no of ways = $5! = 120$

11) let A=1 E=2 C=3 L=4 N=5

∴ ANGLE ≡ 15342

the next line after all numbers starting with 1
and 2nd no 1,2,3,4
= $3 \times 3! = 18$

∴ AFTER THESE

NUMBERS ARE 15234
15243
15324
15342

i.e 4 more

∴ number is 12

∴ ANGLE IS IN 22nd place

1 mark for program
to answer.

2

13a

let $u = n^{n-1}x$ $v = nx$

$u' = (n-1)n^{n-2}x + n^{n-1} = n^{n-2}(n-1)x + n^{n-1}$

$\therefore I_1 = \int_0^{\pi} [-\cos 2n^{n-2}x] + (n-1) \int_0^{\pi} n^{n-2} \cos 2x dx$

$= 0 + (n-1) \int_0^{\pi} n^{n-2} (1 - n^2x) dx$

$= (n-1) \int_0^{\pi} (n^{n-2} - n^n x) dx$

$= (n-1) I_{n-2} - (n-1) I_n$

$I_n = \frac{n-1}{n} I_{n-2}$

$I_5 = \frac{4}{3} I_3$

$I_3 = \frac{2}{3} I_1$

$I_1 = \int_0^{\pi} nx dx = 2$

$\therefore I_5 = \frac{4}{3} \cdot \frac{2}{3} \cdot 2 = \frac{16}{15}$

b)

$P(3) = 3$

$P(7) = 5$

$P(x) = a(x)(x-3)(x-7) + ax + b$

$P(3) = 0 + 3a + b = 3$

$P(7) = 0 + 7a + b = 5$

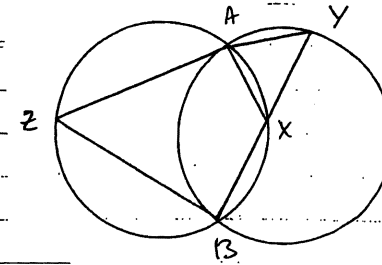
$\therefore 4a = 2$

$a = \frac{1}{2}$

$b = \frac{3}{2}$

$\therefore P(x) = \frac{x}{2} + \frac{3}{2}$

c)



Construct AZ and ZB where Z is on circle

$\angle AZB = \angle AXY$ (EXTERIOR ANGLE OF A

CYCLIC QUAD = THE INTERIOR OPPOSITE)

$\angle AZB = \angle AYB$ (ANGLES SUBTENDED BY

EQUAL ARCS IN CIRCLES OF EQUAL RADI)

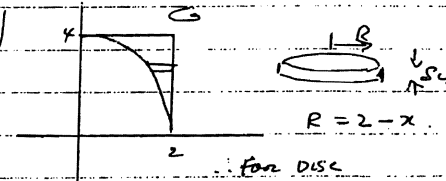
$\therefore \angle AXY = \angle AYB$

$\therefore \triangle AXY$ IS ISOSCELES

$\therefore AX = AY$ (SIDES OPPOSITE EQUAL

ANGLES IN ISOSCELES TRIANGLE)

d)



$\delta V = \pi (2-x)^2 \delta y$

$= \pi (4 - 4x + x^2) \delta x$

now $x^2 = 4 - y$

$\therefore \delta V = \pi (4 - 4\sqrt{4-y} + 4 - y)$

$= \pi (8 - y - 4\sqrt{4-y})$

$V = \int_0^4 \delta V$

$= \pi \int_0^4 (8 - y - 4\sqrt{4-y}) dy$

$= \left[8y - \frac{y^2}{2} + \frac{8}{3} (4-y)^{3/2} \right]_0^4$

$= \frac{8\pi}{3} \text{ units}^3$

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$$i) \quad ma = mg - \frac{mv^2}{100} \quad \downarrow \quad \uparrow \quad \frac{mv^2}{100}$$

$$\therefore a = \ddot{x} = g - \frac{v^2}{100} \quad \downarrow$$

$$ii) \quad \text{TV when } a=0$$

$$\therefore \frac{v^2}{100} = g$$

$$\therefore V = 10\sqrt{g} \text{ m s}^{-1} \quad \downarrow$$

$$iii) \quad v \frac{dv}{dx} = g - \frac{v^2}{100}$$

$$\int \frac{100v}{100g - v^2} dv = \int dx \quad V^2 = 100g$$

$$\therefore x = -50 \ln(V^2 - v^2) + C \quad \downarrow$$

$$\text{at } t=0 \quad v=0$$

$$\therefore C = 50 \ln V^2$$

$$\therefore -\frac{x}{50} = \ln \left(\frac{V^2 - v^2}{V^2} \right)$$

$$\frac{V^2 - v^2}{V^2} = e^{-\frac{x}{50}} \quad \downarrow$$

$$V^2 - v^2 = V^2 e^{-\frac{x}{50}}$$

$$\therefore v^2 = V^2 - V^2 e^{-\frac{x}{50}}$$

$$v^2 = V^2 \left(1 - e^{-\frac{x}{50}} \right) \quad \downarrow$$

$$iv) \quad \text{subst } v = \frac{V}{2} \text{ into } v^2$$

$$\therefore \frac{V^2}{4} = V^2 \left(1 - e^{-\frac{x}{50}} \right) \quad \downarrow$$

$$\therefore e^{-\frac{x}{50}} = \frac{3}{4}$$

$$\therefore -\frac{x}{50} = \ln \frac{3}{4}$$

$$\therefore x = 50 \ln \frac{4}{3} \quad \downarrow$$

$$v) \quad \text{subst } x=50 \text{ into } v^2 \quad \downarrow$$

$$\therefore v^2 = V^2 (1 - e^{-1})$$

$$\therefore v = V \sqrt{\frac{e-1}{e}} \quad \downarrow$$

$$vi) \quad v_1^2 = V^2 \left(1 - e^{-\frac{d}{50}} \right) \quad 2=v_1 \quad x=d$$

$$v_2^2 = V^2 \left(1 - e^{-\frac{2d}{50}} \right) \quad v=v_2 \quad x=2d$$

$$= V^2 \left(1 - e^{-\frac{2d}{50}} \right)$$

$$\therefore \frac{v_2^2}{v_1^2} = \frac{V^2 \left(1 - e^{-\frac{2d}{50}} \right)}{V^2 \left(1 - e^{-\frac{d}{50}} \right)} \quad \downarrow$$

$$\therefore v_2^2 = v_1^2 \left(1 + e^{-\frac{d}{50}} \right)$$

$$\text{now } \frac{v_2^2}{v_1^2} = 1 - e^{-\frac{d}{50}}$$

$$\therefore e^{-\frac{d}{50}} = 1 - \frac{v_2^2}{v_1^2}$$

$$\therefore v_2^2 = v_1^2 \left(1 + 1 - \frac{v_2^2}{v_1^2} \right) \quad \downarrow$$

$$= v_1^2 \left(2 - \frac{v_2^2}{v_1^2} \right) \quad \downarrow$$

$$(4b) \quad P(x) = x^4 - 5x^3 - 9x^2 + ax + b$$

$$P'(x) = 4x^3 - 15x^2 - 18x + a$$

$$P''(x) = 12x^2 - 30x - 18 \quad \downarrow$$

$$\text{for triple root } P''(x) = 0$$

$$\therefore 6(2x+1)(x-3) = 0$$

$$\therefore x=3 \text{ is triple root} \quad \downarrow$$

$$\text{now } 27 = 3+3+3+\beta = 5 \quad \therefore \beta = -4$$

$$2^3 \beta = b$$

$$\therefore 3^3 \cdot -4 = b$$

$$\therefore b = -108 \quad \downarrow$$

$$15a) \quad \text{test for } n=1$$

$$\text{LHS} = (1+x)^1 - 1$$

$$= x \text{ which is divisible by } x$$

$$\therefore \text{true for } n=1$$

$$\text{assume that } (1+x)^k - 1 \text{ is divisible by } x \text{ for } n=k$$

$$\text{i.e. assume } (1+x)^k - 1 = x P(x) \text{ where } P(x) \text{ is a polynomial}$$

$$\text{now prove that } (1+x)^{k+1} - 1 \text{ is divisible by } x$$

$$(1+x)^{k+1} - 1 = (1+x)(1+x)^k - 1$$

$$= (1+x)^k + x(1+x)^k - 1$$

$$= [(1+x)^k - 1] + x(1+x)^k$$

$$\text{as } [(1+x)^k - 1] \text{ is divisible by } x \text{ by assumption}$$

$$\text{and } x(1+x)^k \text{ is " " " "}$$

$$\text{the } (1+x)^{k+1} - 1 \text{ is divisible by } x$$

$$\therefore \text{true for } n=k+1$$

$$\text{Since the result is true for } n=1$$

$$\text{from assumption it is true for } n=k+1=2$$

$$\text{and then for } n=3 \text{ and so on for all } n \geq 1$$

- Marking: 4 components
- test $n=1$
 - uses assumption
 - determines divisibility
 - correct structure
- All 4 correct 3 marks
 3 correct 2 marks
 2 correct 1 mark
 0 otherwise

15a.)

$$\begin{aligned} & 35^n - 7^n - 5^n + 1 \\ &= 7^n 5^n - 7^n - 5^n + 1 \\ &= 7^n (5^n - 1) - (5^n - 1) \\ &= (7^n - 1)(5^n - 1). \end{aligned}$$

for $(7^n - 1)$ let $n = 6$

$((1+6)^n - 1)$ is divisible by 6 from i)
 when 5^n is divisible by 4 from ii)

$(7^n - 1)(5^n - 1)$ is divisible by $6 \times 4 = 24$.

15b)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 4 \cos x}$$

let $t = \tan \frac{x}{2}$

$$dx = \frac{2dt}{1+t^2}$$

$$I = \int_0^1 \frac{dx}{5 + \frac{4(1-t^2)}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

when $x = \frac{\pi}{2}$ $t = 1$

$x = 0$ $t = 0$

$$= \int_0^1 \frac{2dt}{5 + t^2 + 4 - 4t^2}$$

$$= \int_0^1 \frac{2}{9 + t^2} dt$$

$$= \frac{2}{3} \ln \frac{t}{3}$$

$$= \frac{2}{3} \ln \frac{1}{3} - \frac{2}{3} \ln 0$$

$$= \frac{2}{3} \ln \frac{1}{3}$$

15c) i) $x^2 = 4u$

$y^2 = 4u$

$\frac{x^4}{16a^2} = 4u$

$x^4 - 64a^2u = 0$

$x(x^3 - 64u) = 0$

$\therefore x = 0 \text{ or } 4u$

$y = 0 \text{ or } 4u$

$\therefore A(0,0) \quad B(4a,4a)$

ii)



$CD = 2\sqrt{au} - \frac{x^2}{4u}$ (ordinate or semi C)

$\therefore A_{\text{area}} = \frac{\pi r^2}{2}$

where $r = \frac{CD}{2}$

$\therefore A = \frac{\pi}{2} \left(\sqrt{au} - \frac{x^2}{8u} \right)^2$

iii)

$\delta V = A \delta x$

$\delta V = \frac{\pi}{2} \left(\sqrt{au} - \frac{x^2}{8u} \right)^2 \delta x$

$\therefore V = \sum_0^{4a} \delta V$

$= \frac{\pi}{2} \int_0^{4a} \left(\sqrt{au} - \frac{x^2}{8u} \right)^2 dx$

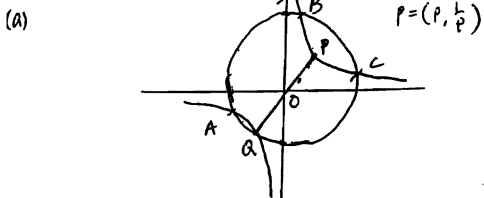
$= \frac{\pi}{2} \int_0^{4a} \left(au - \frac{x^2}{4a^{1/2}} + \frac{x^4}{64a^2} \right) dx$

$= \frac{\pi}{2} \left[\frac{au^2}{2} - \frac{x^{3/2}}{14a^{1/2}} + \frac{x^5}{5 \times 64a^2} \right]_0^{4a}$

$= \frac{\pi}{2} \left[\frac{8a^3}{2} - \frac{64 \cdot 2a^{3/2}}{14} + \frac{4^5 \times 4^2 a^2}{5 \times 64} \right] = 0$

$= \frac{\pi}{2} a^3 \left[8 - \frac{64}{7} + \frac{16}{5} \right]$

$= \frac{36}{35} \pi a^3$



1) $y = \frac{1}{x}$ is an odd function.
It has 'half turn' symmetry.

$$\therefore OP = OQ$$

$$Q = (-p, -\frac{1}{p})$$

$$r = PQ = 2OP$$

$$= 2\sqrt{p^2 + \frac{1}{p^2}}$$

$$= 2\sqrt{\frac{p^4 + 1}{p^2}}$$

$$r^2 = 4\frac{(p^4 + 1)}{p^2}$$

Equation of circle centre P, radius = r is

$$(x-p)^2 + (y-\frac{1}{p})^2 = 4\frac{(p^4 + 1)}{p^2} \quad \text{1 mark}$$

If a point (x, y) on the circle also lies on the hyperbola $y = \frac{1}{x}$ then $(x, y) = (t, \frac{1}{t})$, for some values of t and $(t, \frac{1}{t})$ also satisfies the equation of the circle.

$$\therefore (t-p)^2 + (\frac{1}{t} - \frac{1}{p})^2 = 4\frac{(p^4 + 1)}{p^2} \quad \text{1 mark}$$

$$t^2 + p^2 - 2tp + \frac{1}{t^2} + \frac{1}{p^2} - \frac{2}{tp} = \frac{4(p^4 + 1)}{p^2}$$

$$t^2 - 2tp + (p^2 + \frac{1}{p^2}) - \frac{2}{tp} + \frac{1}{t^2} = \frac{4(p^4 + 1)}{p^2}$$

$$t^2 - 2tp - 3\frac{(p^4 + 1)}{p^2} - \frac{2}{tp} + \frac{1}{t^2} = 0$$

$$p^2, p^2t^4 - 2p^3t^3 - 3(p^4 + 1)t^2 - 2pt + p^2 = 0 \quad \text{1 mark}$$

i) $t_A + t_B + t_C + t_Q = \frac{2p^3}{p^2} \quad \text{1 mark}$

$$t_A + t_B + t_C - p = 2p$$

$$t_A + t_B + t_C = 3p \quad \text{1 mark}$$

ii) LHS = $(1 + \frac{e^{i\sin\theta}}{\cos\theta})^n + (1 - \frac{e^{i\sin\theta}}{\cos\theta})^n$

$$= \frac{(\cos\theta + i\sin\theta)^n}{\cos^n\theta} + \frac{(\cos\theta - i\sin\theta)^n}{\cos^n\theta} \quad \text{1 mark}$$

$$= \frac{\cos^n\theta + i\sin^n\theta}{\cos^n\theta} + \frac{(\cos\theta - i\sin\theta)^n}{\cos^n\theta}$$

$$= \frac{\cos^n\theta + i\sin^n\theta}{\cos^n\theta} + \frac{\cos^n\theta - i\sin^n\theta}{\cos^n\theta}$$

$$= \frac{\cos^n\theta + i\sin^n\theta + \cos^n\theta - i\sin^n\theta}{\cos^n\theta}$$

$$= \frac{2\cos^n\theta}{\cos^n\theta} \quad \cos\theta \neq 0 \quad \text{1 mark}$$

(ii) If $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
the range of $\tan\theta$ is all real values
Hence any imaginary number z
can be written as $i\tan\theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$(1+z)^4 + (1-z)^4 = 0$$

$$\Leftrightarrow (1+i\tan\theta)^4 + (1-i\tan\theta)^4 = 0$$

$$\Leftrightarrow \frac{2\cos^4\theta}{\cos^4\theta} = 0 \quad \text{1 mark}$$

if $\cos^4\theta = 0$
 $4\theta = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}$ $2\pi < 4\theta < 2\pi$
 $\theta = \pm\frac{\pi}{8}, \pm\frac{3\pi}{8}$ 1 mark

$$\therefore z = i\tan\pm\frac{\pi}{8}, i\tan\pm\frac{3\pi}{8}$$

$$\tan\pm\frac{\pi}{8} = -\tan\pm\frac{\pi}{8}$$

$$\tan\pm\frac{3\pi}{8} = -\tan\pm\frac{3\pi}{8}$$

$$\therefore z = \pm i\tan\frac{\pi}{8}, \pm i\tan\frac{3\pi}{8} \quad \text{1 mark}$$

(ii) $V_k = \frac{1}{2k+1} + \frac{1}{2k+2} + \dots + \frac{1}{3k}$

$$2k < 2k+1 < 2k+2 < \dots < 3k$$

$$\Rightarrow \frac{1}{3k} < \frac{1}{2k+1} < \dots < \frac{1}{2k+2} < \frac{1}{2k+1} < \frac{1}{2k}$$

$$\therefore V_k < \frac{1}{2k} + \frac{1}{2k} + \dots + \frac{1}{2k} \quad (k \text{ terms})$$

$$= \frac{1}{2k} \times k$$

$$= \frac{1}{2} \quad \text{1 mark}$$

(ii) $0 < p < x < p+1$

$$\Rightarrow \frac{1}{p+1} < \frac{1}{x} < \frac{1}{p}$$

$$\therefore \int_p^{p+1} \frac{1}{x} dx < \int_p^{p+1} \frac{1}{x} dx < \int_p^{p+1} \frac{1}{p} dx$$

$$\left[\frac{x}{p+1}\right]_p^{p+1} < \int_p^{p+1} \frac{1}{x} dx < \left[\frac{x}{p}\right]_p^{p+1}$$

$$\frac{1}{p+1} < \int_p^{p+1} \frac{1}{x} dx < \frac{1}{p} \quad \text{1 mark}$$

(iii) Sub. $p = 2k+1$ into $\int_p^{p+1} \frac{dx}{x} < \frac{1}{p}$

We get $\int_{2k+1}^{2k+2} \frac{dx}{x} < \frac{1}{2k+1}$

Sub. $p = 2k+2$ $\int_{2k+2}^{2k+3} \frac{dx}{x} < \frac{1}{2k+2}$

Sub. $p = 3k$ $\int_{3k}^{3k+1} \frac{dx}{x} < \frac{1}{3k}$

Adding $\int_{2k+1}^{3k+1} \frac{dx}{x} < \frac{1}{2k+1} + \dots + \frac{1}{3k}$

$$\therefore \int_{2k+1}^{3k+1} \frac{dx}{x} < V_k \quad \text{1 mark}$$

Sub. $p = 2k$ into $\frac{1}{p+1} < \int_p^{p+1} \frac{dx}{x}$

We get $\frac{1}{2k+1} < \int_{2k}^{2k+1} \frac{dx}{x}$

Sub $p = 2k+1$, $\frac{1}{2k+2} < \int_{2k+1}^{2k+2} \frac{dx}{x}$

Sub. $p = 3k-1$, $\frac{1}{3k} < \int_{3k-1}^{3k} \frac{dx}{x}$

Adding, $\frac{1}{2k+1} + \frac{1}{2k+2} + \dots + \frac{1}{3k} < \int_{2k}^{3k} \frac{dx}{x}$

i.e. $V_k < \int_{2k}^{3k} \frac{dx}{x}$ 1 mark

$$\therefore \int_{2k+1}^{3k+1} \frac{dx}{x} < V_k < \int_{2k}^{3k} \frac{dx}{x}$$

(iv) $\therefore \ln \frac{3k+1}{2k+1} < V_k < \ln \frac{3k}{2k}$

$$\ln \frac{3+\frac{1}{k}}{2+\frac{1}{k}} < V_k < \ln \frac{3}{2}$$

as $k \rightarrow \infty$ $\ln \frac{3+\frac{1}{k}}{2+\frac{1}{k}} \rightarrow \ln \frac{3}{2}$

$\therefore \lim_{k \rightarrow \infty} V_k = \ln \frac{3}{2}$ 1 mark